# Ojasvi Shrivastava

B.A. Data Science · UC Berkeley · Class of 2027

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## EDUCATION

**University of California, Berkeley** — *Berkeley, CA*
*B.A., Data Science* — *Expected May 2027*
**Relevant Coursework:** Probability, Linear Algebra, Multivariable Calculus, Algorithms & Data Structures, Machine Learning, Artificial Intelligence

## TECHNICAL SKILLS

- **Languages:** Python, C++, Java, TypeScript, SQL
- **Machine Learning / Data:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Statsmodels, Monte Carlo / GBM, K-Means, Multivariate Regression, Spatial Econometrics, GeoPandas, Matplotlib, Plotly, Tableau
- **Web Development:** Next.js, React, TypeScript, Tailwind CSS, Recharts, Framer Motion, Vercel
- **Tools:** Git, Jupyter, Linux, ML Pipelines, Vectorization & Numerical Optimization, Algorithm Design

## EXPERIENCE

**Private Mathematics Tutor** — *2024 – 2025*

- Designed structured first principles problem-solving frameworks for calculus, linear algebra, and statistics, diagnosing each student's binding constraint and resequencing material to attack it directly, yielding an **average 60% exam score improvement** across tutees.

## PROJECTS

**FinSight** — *Full-Stack Wealth & Retirement Simulator* — *2025 – Present*
*TypeScript, Next.js, Monte Carlo / GBM, localStorage* — *Live Site GitHub*

- Engineered a vectorized stochastic engine in TypeScript running a **3,000-trial Geometric Brownian Motion Monte Carlo simulation** with seeded PRNG for deterministic P10/P50/P90 percentile retirement forecasting.
- Built a rule based **AI insight engine** that reads the user's actual savings rate, portfolio P&L, emergency fund gap, and sector risk to fire specific, numbered recommendations rather than generic financial tips.
- Shipped a cash flow tracker with category icons and repeat-last shortcut, an emergency fund ring chart, a live market sentiment gauge, and a **six-article financial education curriculum** with interactive quizzes all **zero-backend**, localStorage only, no account required.

**GIStice League** — *Spatial Econometrics for Food-Access Inequity* — *Berkeley Open Project*
*Machine Learning Team | Python, PyTorch, Scikit-learn, GeoPandas* — *Live Site GitHub*

- Engineered spatial econometric features, including **K-Means cluster identifiers and neighborhood grocery density gradients** into a PyTorch multivariate regression, achieving an **18% $R^2$ accuracy gain** over naive distance only baselines across 10,000+ U.S. Census tracts.
- Built a reproducible pipeline to ingest Census shapefiles and render interactive **Leaflet + GeoPandas choropleths** with 20+ live ML features and real-time $R^2$ / RMSE updates; adopted by community advocates for targeted food infrastructure policy.

**Interactive ML Demo** — *Browser-Native Algorithm Visualization* — *Public Portfolio*
*TypeScript, HTML5 Canvas, Closed-Form Linear Algebra* — *Live Demo*

- Programmed a real-time, browser-native **4-feature multivariate housing regression** entirely from scratch in TypeScript closed-form solver and live prediction updates run with no backend.